N=2 $a=\frac{\pi}{2}$, $b=2\pi$, $h=2\pi-\frac{\pi}{2}=\frac{3\pi}{2}$, $x_0=\frac{\pi}{2}$, $x_1=2\pi$, $y_0=\frac{\pi}{2}$

$f(x_0,y_0)=\frac{y_0}{x_0}+x_0\cos x_0$

$f\left(\frac{\pi}{2},\frac{\pi}{2}\right)=\frac{\pi/2}{\pi/2}+\frac{\pi}{2}\cos\frac{\pi}{2}=1+0=1$

пред: (0)

$y_1=y_0+hf(x_0,y_0)=\frac{\pi}{2}+\frac{3\pi}{2}=2\pi\approx 6{,}283$

корр.:

$y_1^{(1)}=y_0+\frac{h}{2}\left(f(x_0,y_0)+f(x_1,y_1^{(0)})\right)\in$

$f(x_1,y_1^{(0)})=\frac{2\pi}{2\pi}+2\pi\cos(2\pi)=1+2\pi\approx 7{,}283$

$\in \frac{\pi}{2}+\frac{3\pi}{4}(1+7{,}283)=1{,}571+2{,}356\cdot 8{,}283\approx 21{,}091$

$f(x_1,y_1^{(1)})=\frac{21{,}091}{2\pi}+2\pi\approx 9{,}64$

$y_1^{(2)}=\frac{\pi}{2}+\frac{3\pi}{4}(1+9{,}64)\approx 26{,}64$

$f(x_1,y_1^{(2)})=\frac{26{,}64}{2\pi}+2\pi\approx 10{,}523$

$y_1^{(3)}=1{,}571+2{,}356(1+10{,}523)\approx 28{,}721$

$y_1^{(4)}\approx 29{,}501$

$\Rightarrow |y_1^{(2)}-y_1^{(1)}|=5{,}55$

$\Rightarrow |y_1^{(3)}-y_1^{(2)}|\approx 2{,}081$

$\Rightarrow |y_1^{(4)}-y_1^{(3)}|\approx 0{,}78$